

The Gamma distribution is specified by two parameters positive α, β . Its MGF is $M(t) = \frac{1}{(1-\beta t)^\alpha}$.

1. Suppose X is a random variable with Gamma distribution with $\alpha = 3, \beta = 2$. Use the MGF to compute expectation $E[X]$ and variance $\text{Var}[X]$.

$$(1-\beta t)^{-\alpha} \quad \alpha\beta(1-\beta t)^{-\alpha-1}$$

$$E[X] = 6 \cdot 1^{-4} = 6$$

$$\alpha(\alpha-1)\beta^2(1-\beta t)^{-\alpha-2}$$

3 - 4 2 - 2 1 - 5

$$E[X^2] = 48$$

$$\text{Var}[X] = 48 - 6^2 = 12$$

2.

3. Compute the MGF of sample mean $\bar{X}$ of a random sample of size n from a Gamma population with $\alpha = 3, \beta = 2$. Is $\bar{X}$ a Gamma random variable? Why or why not.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$M_{\bar{X}}(t) = M_X\left(\frac{t}{n}\right)^n = (1-2\frac{t}{n})^{-3n} = \frac{1}{(1-\frac{2}{n}t)^{3n}}$$

$\bar{X}$ is a gamma random variable with $\alpha = 3n$ and $\beta = \frac{2}{n}$

4. Continuing previous question: What is the mean and variance of $\bar{X}$

$$3n \cdot \frac{2}{n} \cdot 1^{-3n-1} = 6 \quad E[\bar{X}] = E[X] = 6$$

$$3n(3n-1)\frac{2}{n}\left(-\frac{2}{n}\right)1^{-3n-2} = 6(3n-1)\left(-\frac{2}{n}\right) = \frac{12}{n}(3n+1) = 36 + \frac{12}{n} = E[\bar{X}^2]$$

$$\text{Var}[\bar{X}] = E[\bar{X}^2] - E[\bar{X}]^2 = 36 + \frac{12}{n} - 36 = \frac{12}{n}$$